

Technology Stack

Date	29 September 2026
Team ID	6ab4fddea238dc999eb665e9
Project Name	PocketSmart AI: Your Smart Budget & Recommendation Assistant
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript, Jinja2 Templates	Provides the web interface for the Home, Party and Jewelry planners. Jinja2 connects dynamic FastAPI data to the pages, while HTML/CSS/JavaScript keep the interface lightweight and maintainable.

2	Backend / Server-Side	Python, FastAPI, Uvicorn	FastAPI provides the API and routing layer, Python supports authentication, validation and business logic, and Uvicorn serves the application through ASGI.
3	Database / Data Storage	SQLite	SQLite provides lightweight persistent storage for users, session/recommendation data and application history without requiring a separate database server for the student project.
4	Cloud / Hosting / Deployment	Uvicorn / FastAPI deployment	The prototype runs through Uvicorn and follows a standard FastAPI application structure, keeping deployment straightforward without changing the core application architecture.
5	Version Control & CI/CD	Git + GitHub + pytest	Git and GitHub support source control, collaboration and versioned submission. pytest provides repeatable automated tests for core application behavior; a separate CI pipeline is not required for the current prototype.

6	Third-Party APIs / Other Tools	Gemini API (google-genai)	Gemini is used for AI-powered personalized recommendations and structured output generation, with multimodal image analysis available for the Jewelry planner when configured. API credentials are supplied through environment variables.
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